

MARS: A Safety-Filtered Multi-Robot Area-Coverage Testbed with Classical and MAPPO Controllers

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Abstract—We describe MARS, an open ROS 2 Jazzy / Gazebo Harmonic testbed for cooperative area coverage by a small team of differential-drive robots, and report what it showed. Three controllers share one safety stack (two fixed override rules followed by a soft control-barrier-function filter): a classical planner (frontier selection, A* routing, Voronoi partitioning with consensus-based auction claims exchanged within a 3 m radio range, and a behavior tree), a random walk, and a MAPPO policy with a permutation-invariant neighbor encoder and a centralized critic. Given enough steps, the classical planner covered 100.0% of the sensed, non-obstacle area in the warehouse, 98.2% in the office, 85.7% in the depot, 100.0% in the cafe and 25.0% in the maze, which is a property of those worlds and step budgets as much as of the controller. In a robot-count sweep (one run per cell) coverage generally rose with team size, and robots touched in 3 of 20 configurations. Trained MAPPO baselines covered less than the classical planner in every world of an exploratory single-episode matrix; a reward ablation with three training seeds per arm was inconclusive. We also report three limits of the testbed itself: evaluation outcomes vary between repeated runs of an identical checkpoint, the safety margin is soft, and we did not test the architecture’s claimed ability to transfer to other team sizes. All data and code are archived with a DOI.

Index Terms—multi-robot systems, area coverage, control barrier functions, MAPPO, reproducibility

I. INTRODUCTION

Cooperative area coverage, in which a team of robots jointly sweeps an unknown environment, underlies search, inspection and monitoring. Evaluating coordination methods for it needs a simulator that models range sensing and contact, and results that others can reproduce. MARS is a testbed built for that purpose. It wraps Gazebo in a multi-agent environment, provides a classical decentralized controller and a learned MAPPO baseline behind the same safety stack, and archives per-run results.

Contributions.

- A documented testbed with five worlds, three controllers and a common safety stack, described here as implemented (Sec. III).
- Coverage ceilings of the classical controller in the five worlds, a robot-count sweep, and a comparison with trained MAPPO baselines, all from archived runs (Sec. IV).
- An account of the testbed’s limits: non-deterministic evaluation, a soft safety margin, and claims that remain untested (Sec. V).

We make no claim that the learned policy is competitive, that the safety filter gives a formal guarantee, or that the policy transfers to other team sizes.

II. RELATED WORK

Frontier-based exploration [1] and its multi-robot extensions [2], [3] are the classical baseline for this task; the consensus-based bundle algorithm [4] supplies the auction mechanism we use to resolve duplicate target claims. MAPPO [5] is a strong baseline for cooperative multi-agent learning; we train it with RLlib [6]. The actor’s neighbor encoder is a mean-pooled embedding in the style of Deep Sets [7]; it contains no message passing, so we do not call it a graph neural network. Control barrier functions [8] give a quadratic-program safety filter, and barrier-function filters have been placed around learned policies [9]. We solve ours with OSQP [10] and add a slack variable, which makes it soft (Sec. III-C). A companion study [11] examines the MAPPO baseline in depth.

III. THE TESTBED

A. Platform and Metrics

Three TurtleBot3 Waffle robots (differential drive, 24-sector LiDAR, odometry) run in Gazebo Harmonic under ROS 2 Jazzy [12], [13]. The simulator is wrapped as a PettingZoo [14] parallel environment. The workspace is a grid of 0.4 m cells. A cell is marked visited when a robot’s pose falls in it or a LiDAR ray (up to 2.0 m) passes through it as free space. The *area-coverage rate* is

$$\text{ACR} = \frac{|\text{visited} \setminus \text{obstacle}|}{N_{\text{cells}} - |\text{obstacle}|} \times 100, \quad (1)$$

where “obstacle” cells are those the robots have registered as obstacles, so the denominator depends on what has been discovered. A *wall contact* is counted when the minimum LiDAR range is below 0.14 m (the approximate sensor-to-bumper distance); an *agent contact* when two robots are closer than 0.20 m after the first 15 steps. The agent-contact threshold is smaller than the filter’s agent margin (Sec. III-C), so a zero count does not show that the margin held.

B. Observation and Action

Each robot observes a 46-vector: 24 LiDAR sector ranges, goal distance and bearing, linear and angular velocity, and nine neighbor slots of (distance, bearing), padded with (10.0, 0.0). Actions are (v, ω) with $|v| \leq 0.22$ m/s and $|\omega| \leq 1.0$ rad/s. Because the vector has a fixed length, the network accepts up to nine neighbors.

C. Safety Stack

Every controller’s command passes through three stages. (1) *Override rules*. If the minimum LiDAR range over all beams is below 0.22 m and the commanded forward speed is positive, forward speed becomes -0.08 m/s; if another robot is within 0.45 m and within 72° of the heading, forward speed becomes -0.05 m/s. (2) *Soft CBF*. A quadratic program (OSQP) solves

$$\begin{aligned} \min_{v, \omega, \delta} & (v - v_{\text{nom}})^2 + 0.05(\omega - \omega_{\text{nom}})^2 + P\delta^2 \\ \text{s.t.} & A_k v + B_k \omega + \delta \geq -\gamma h_k, \end{aligned} \quad (2)$$

with $0 \leq \delta \leq 10$, $P = 1000$ and $\gamma = 2.0$. Barrier terms come from LiDAR returns within 0.6 m, evaluated at two lookahead points (± 0.12 m on the heading axis) with obstacle radius 0.20 m, and from neighbors within 1.2 m with agent radius 0.45 m. The neighbor term treats the neighbor’s velocity as bounded and unmodeled. If the QP is infeasible the filter returns $(\min(0, v_{\text{nom}}), \omega_{\text{nom}})$. (3) *Front safeguard*. Forward speed is set to $\min(0, v)$ if any of the five forward-cone beams reads below 0.24 m. Because of the slack δ , the radii are soft constraints: we claim no formal guarantee and report behaviour only.

D. Classical Controller

Each robot keeps its own visited and obstacle grids, built from its LiDAR. When two robots are within 3.0 m they merge their visited grids by bitwise OR, a range-limited approximation of gossip consensus; a global grid is kept only for monitoring and for computing ACR. Frontier cells are the nearest unvisited candidates inside the robot’s own Voronoi cell relative to sensed neighbors, duplicate claims are resolved by a consensus-based auction among peers in range, and a stale claim from a peer that has gone out of range expires after 30 steps. Routes use 8-connected A* on an obstacle-inflated grid. A behavior tree selects, in priority order, mission-complete standby or perimeter patrol, an escape maneuver ($v = -0.12$ m/s, $\omega = \pm 0.6$ rad/s) when a robot has not moved for 30 steps, frontier exploration, and a boundary-aware wander. Unreachable targets are blacklisted. A correlative scan matcher corrects odometry drift.

E. MAPPO Baseline

A single policy is shared by all robots. Each neighbor’s (distance, bearing) pair is embedded by a two-layer ReLU network ($2 \rightarrow 16 \rightarrow 16$), masked and mean-pooled, concatenated with the 28 ego features and passed through a two-layer Tanh actor ($44 \rightarrow 64 \rightarrow 64$). A centralized critic ($76 \rightarrow 64 \rightarrow 64 \rightarrow 1$, Tanh) takes the same 44 features

TABLE I
CLASSICAL CONTROLLER, THREE ROBOTS, ONE LONG RUN PER WORLD.
CONTACTS ARE WALL / AGENT COUNTS OVER THE WHOLE RUN;
CLEARANCE IS THE MINIMUM INTER-ROBOT DISTANCE.

World	Cells	Steps	ACR	Plateau	Dist.	Wall	Agent	Clr. (m)
warehouse	9,375	9,050	100.0	7,660	1263	17,018	0	0.81
office	3,500	3,500	98.2	3,466	377	0	0	0.38
depot	3,000	3,500	85.7	2,906	180	477	0	0.55
cafe	800	1,200	100.0	397	82	738	0	0.46
maze	4,800	4,000	25.0	3,961	316	1,108	0	0.26

plus a mean-pooled embedding ($48 \rightarrow 32 \rightarrow 32$) of the other agents’ observation–action pairs. The mean-pooled embedding is invariant to neighbor order and padding position, which `tests/test_permutation_invariance.py` checks directly. Training uses RLlib PPO [15] with a single in-process rollout worker, train batch 700, 90-step episodes, minibatch 64, 5 epochs, learning rate 10^{-4} , clip 0.2, $\gamma = 0.99$, GAE $\lambda = 0.95$ and entropy coefficient 0.01. The per-step reward is

$$\begin{aligned} r_i = & -0.01 + 5.0 \Delta d_i + 4.0 n_{\text{new}} + 20 \mathbb{I}[\text{goal}] \\ & - 8.0 \mathbb{I}[\text{wall}] - \rho_i, \end{aligned} \quad (3)$$

where Δd_i is the reduction in goal distance, n_{new} the number of cells newly visited by any robot this step, and $\rho_i = 0.5$ if another robot is within 0.5 m plus 10 within 0.20 m. The goal is a random point from a fixed per-world list of safe positions, reassigned on arrival; it is less informed than the classical controller’s map-aware frontier targets. Evaluation uses greedy actions.

IV. RESULTS

Protocol and caveats. Unless stated, each cell is one episode. We found that repeating an evaluation of an identical checkpoint and seed gives different outcomes (Sec. V), so single-episode numbers carry unquantified run-to-run variance; we treat them as descriptive. Raw per-run records are archived (Data availability).

A. Coverage Ceilings of the Classical Controller

Table I reports one long run per world with three robots. The step budget differs by world and was chosen to see where coverage levels off, so “steps to plateau” is the first sampled step at which ACR came within 0.5 points of its final value; it is not a controller-versus-controller comparison.

In 3 of 5 worlds (warehouse, depot, cafe) ACR reached its plateau before 90% of the run, so those budgets were long enough to see it level off; for the others the final value may still have been rising. The maze’s figure should not be read as a success or a failure by itself: we did not measure how much of its bounding box is traversable, and the ACR denominator only excludes cells where a LiDAR return has registered an obstacle. The warehouse run recorded a very large wall-contact count over its long perimeter patrol after coverage saturated, which the contact metric (sensor range below 0.14 m) makes easy to accumulate.

TABLE II
CLASSICAL CONTROLLER, 200 STEPS, ONE RUN PER CELL: ACR %
(AGENT CONTACTS).

World	$N=2$	$N=3$	$N=5$	$N=8$
warehouse	3.4 (0)	4.8 (0)	5.3 (0)	8.7 (0)
office	5.9 (0)	6.1 (0)	12.7 (0)	46.8 (2)
depot	16.9 (0)	16.6 (0)	31.9 (0)	54.4 (0)
cafe	64.3 (0)	69.3 (0)	66.9 (2)	98.6 (8)
maze	1.1 (0)	3.2 (0)	6.0 (0)	10.0 (0)

TABLE III
ACR % AFTER 150 STEPS, THREE ROBOTS, ONE EPISODE PER CELL.

World	Classical	Random	MAPPO
cafe	57.0	22.6	28.2
warehouse	4.3	2.6	3.3
depot	13.3	6.8	6.7
office	3.8	1.7	3.0
maze	2.7	1.5	2.2

B. Team Size

Table II varies the number of robots from 2 to 8 in each world for 200 steps (one run per cell). Coverage generally rose with team size, though not monotonically (cafe, $N=3 \rightarrow 5$), and the 200-step horizon is far from the plateau in the larger worlds. Agent contacts occurred in 3 of 20 configurations, all at $N \geq 5$ in the two smallest arenas (cafe and office). No other explanation for that pattern was tested.

C. Learned versus Classical

Table III compares the trained MAPPO policy (trained on the cafe, zero-shot elsewhere) with the classical controller and a random walk, one 150-step episode per cell with three robots. The classical controller covered more than MAPPO in all five worlds. This single-episode matrix is exploratory; the companion study [11] reports a 10-episode cafe comparison in which MAPPO’s median coverage (14.5%) was below both the classical controller’s (38.6%) and a random walk’s (29.6%); the per-episode data for that comparison are not archived. Because the controllers differ in information (the classical one builds a map, MAPPO sees only local ranges and a random goal), this compares an engineered system with a minimally trained learned one, not two learned policies.

A reward ablation trained three MAPPO variants for 15 iterations with three training seeds each. Averaged over seeds, coverage was 28.0%, 31.0% and 40.5% for the baseline, a lower collision penalty and a higher discovery bonus. Paired tests across the three seeds did not reach $p < 0.05$ for either variant, so we draw no conclusion about the rewards; see the companion study for the analysis.

D. Safety Observations

Across the 20 sweep runs, the 25 matrix runs and a 27-episode ablation (72 runs), agent contacts were absent in 69 of 72, while coverage ranged from 1.1% to 98.6% (across worlds and horizons), so the contact count did not distinguish weak

from strong controllers. In the office run the minimum inter-robot distance was 0.38 m, below the 0.45 m agent radius, with no contact, which is the behaviour expected of a soft constraint. Two moving-hazard scenarios (head-on at 0.38 m/s and crossing at 0.22 m/s) were run once each without contact; their raw traces are not archived.

V. LIMITATIONS

- **Non-deterministic evaluation.** Three evaluations of one saved checkpoint with the same seed argument gave coverage between 24.6% and 37.8%, and contact counts of 0 to 19; the cause is simulator timing, since evaluation uses greedy actions. Single-episode cells (Tables II and III) are descriptive only.
- **Soft safety margin; no formal guarantee.** See Sec. III-C. We did not measure how often or how much the override rules and filter change each controller’s commands.
- **Transfer to other team sizes is untested.** The encoder accepts up to nine neighbors and is order-invariant, but no trained policy was evaluated with a different number of robots. The sweep uses the classical controller.
- **Ablations not run.** We did not ablate the CBF, the consensus merge, the behavior tree or the scan matcher, so we cannot say how much each contributes.
- **Scope.** Simulation only, one robot model, three robots in the learned experiments, one training world, short training (about 3×10^4 steps), and no comparison with other published coverage planners.

VI. CONCLUSION

MARS provides a documented, archived testbed for safety-filtered multi-robot coverage. Its classical controller reached high coverage in all five worlds given enough steps, coverage generally grew with team size, and a MAPPO baseline underperformed it. The testbed’s own limits matter as much as its numbers: evaluation is not deterministic, the safety margin is soft, and several architectural claims remain untested. We release the code, per-run logs and scripts so that others can check and extend these results.

Data availability.: Code, per-run results and scripts are in the project repository, archived at <https://doi.org/10.5281/zenodo.22758286> (directory checkpoints/).

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